Preface

These lessons were written for those who know little or nothing about modern science. Together they provide a rapid overview of the most fascinating aspects of the great revolution that has occurred in physics in the twentieth century, and of the questions and mysteries which this revolution has opened up. Because science shows us how to better understand the world, but it also reveals to us just how vast is the extent of what is still not known.

The first lesson is dedicated to Albert Einstein's general theory of relativity, 'the most beautiful of theories'. The second to quantum mechanics, where the most baffling aspects of modern physics lurk. The third is dedicated to the cosmos: the architecture of the universe which we inhabit; the fourth to its elementary particles. The fifth deals with quantum gravity: the attempts which are underway to construct a synthesis of the major discoveries of the twentieth century. The sixth is on probability and the heat of black holes. The final section of the book returns to ourselves, and asks how it is possible to think about our existence in the light of the strange world described by physics.

The lessons are expansions of a series of articles published by the author in the Sunday supplement of the Italian newspaper *Il Sole 24 Ore*. I would like to thank in particular Armando Massarenti, who can be credited with opening up the cultural pages of a Sunday paper to science, and allowing light to be thrown on the role of this integral and vital aspect of our culture.